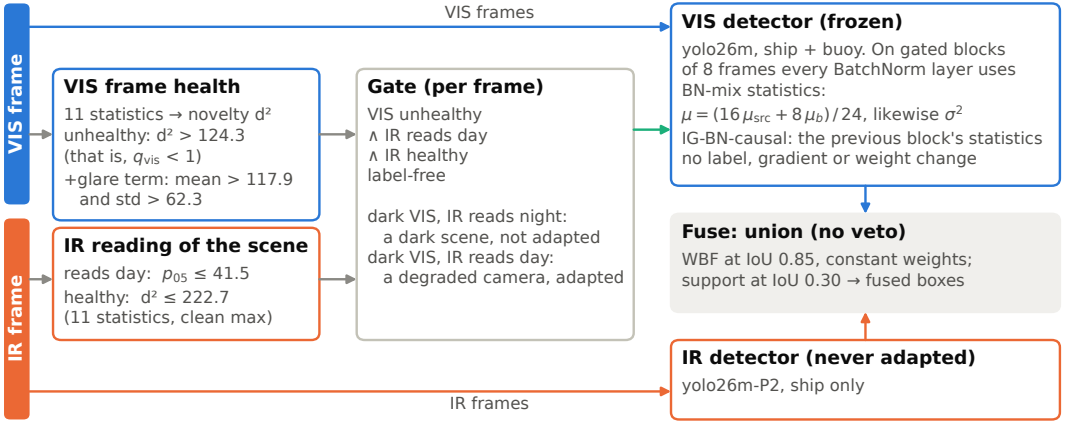
VIS frame
IR frame
VIS frames
VIS frame health
11 statistics → novelty d²
unhealthy: d² > 124.3
(that is, q_{vis} < 1)
+glare term: mean > 117.9
and std > 62.3
IR reading of the scene
reads day: p_{05} ≤ 41.5
healthy: d² ≤ 222.7
(11 statistics, clean max)
Gate (per frame)
VIS unhealthy
∧ IR reads day
∧ IR healthy
label-free
dark VIS, IR reads night:
a dark scene, not adapted
dark VIS, IR reads day:
a degraded camera, adapted
VIS detector (frozen)
yolo26m, ship + buoy. On gated blocks
of 8 frames every BatchNorm layer uses
BN-mix statistics:
$\mu = (16\,\mu_{src} + 8\,\mu_b)/24$, likewise σ^2
IG-BN-causal: the previous block's statistics
no label, gradient or weight change
Fuse: union (no veto)
WBF at IoU 0.85, constant weights;
support at IoU 0.30 → fused boxes
IR detector (never adapted)
yolo26m-P2, ship only
IR frames
Where compute allows (IG-BN+ens):
the same gate and adaptation on all ten VIS views of full_tta (5 checkpoints × raw and flipped)